TEACHING WALLAH

Teaching Exam

Subject – Computer Science

Data Structure and Algorithm

Lecture No -1

By-Satyendra Chaudhary Sir

Topics to be Covered

Topic

→ Data Structure & Algorithm
  ↓
  Searching, Sorting

→ Loops

Data structure
- → Linked list
- → Trees
- → Stack
- → Queues
- → Tables

What is a loop in Python?

A loop in Python is a control flow statement that allows you to execute a piece of code repeatedly until a specific condition is met.

Loops in Python are necessary for operations that require repetitive execution, such as iterating through a Python list of items or doing calculations many times.

Different Types of Loops in Python

To understand the Loop Structures in Python, there are 3 Types of Loops in Python:

- while loop ✓
- for loop ✓
- nested loop

while (True) False

{

while():

{ while(): }

for (.

{

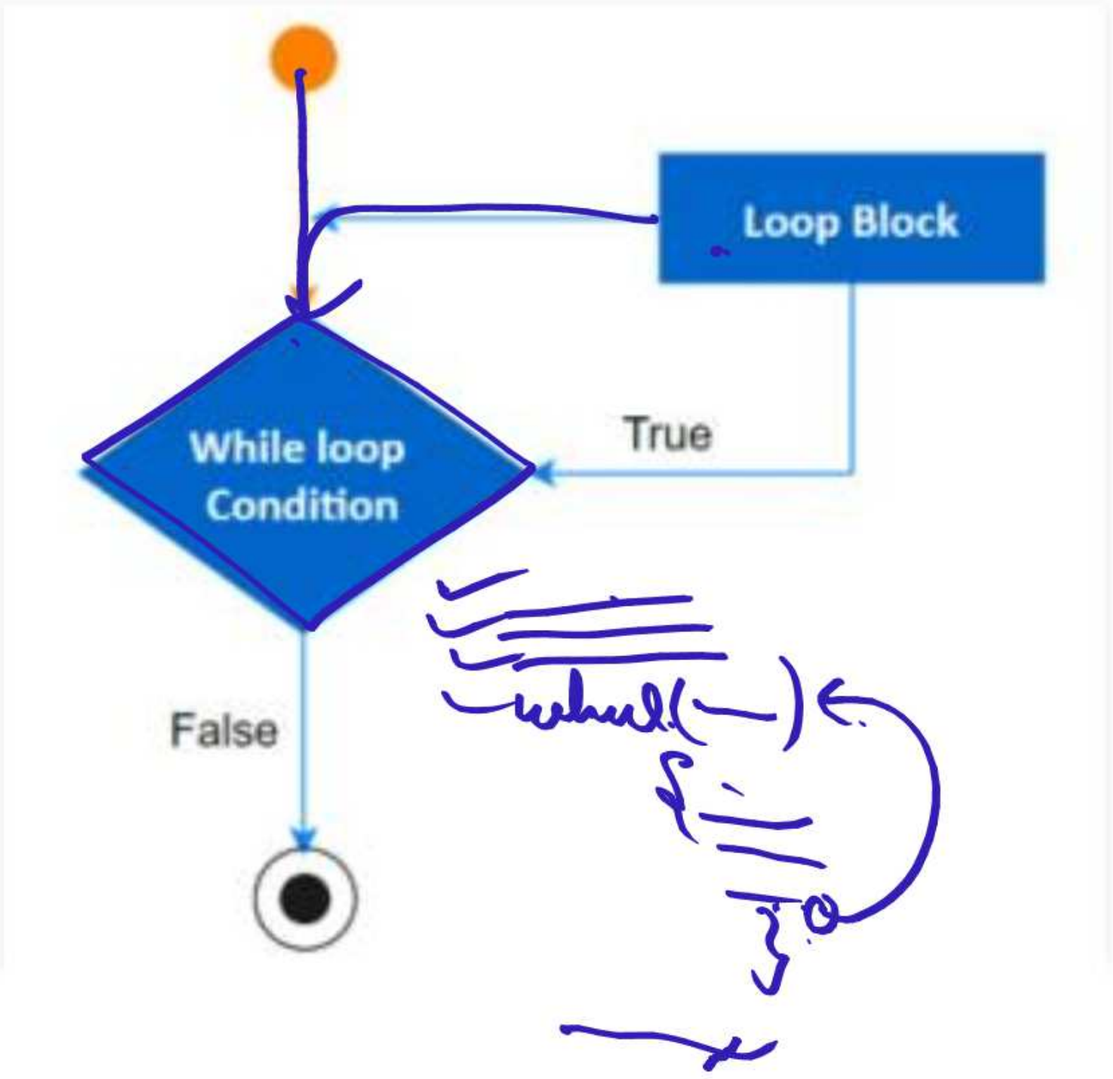
}

Python while loop

A while loop in Python is used to repeat a block of code as long as a certain condition is true.

It keeps running the loop body again and again until the condition becomes false.

This is useful when you don't know in advance how many times the loop should run.



```
count = 0
while (count < 10):
    print ('The count is:', count)
    count = count + 1
print ("Good bye!")
```

A while loop is a control flow statement that allows you to repeatedly run a block of code as long as a given condition is true.

list = [2, 5, 4]

TEACHING WALLAH

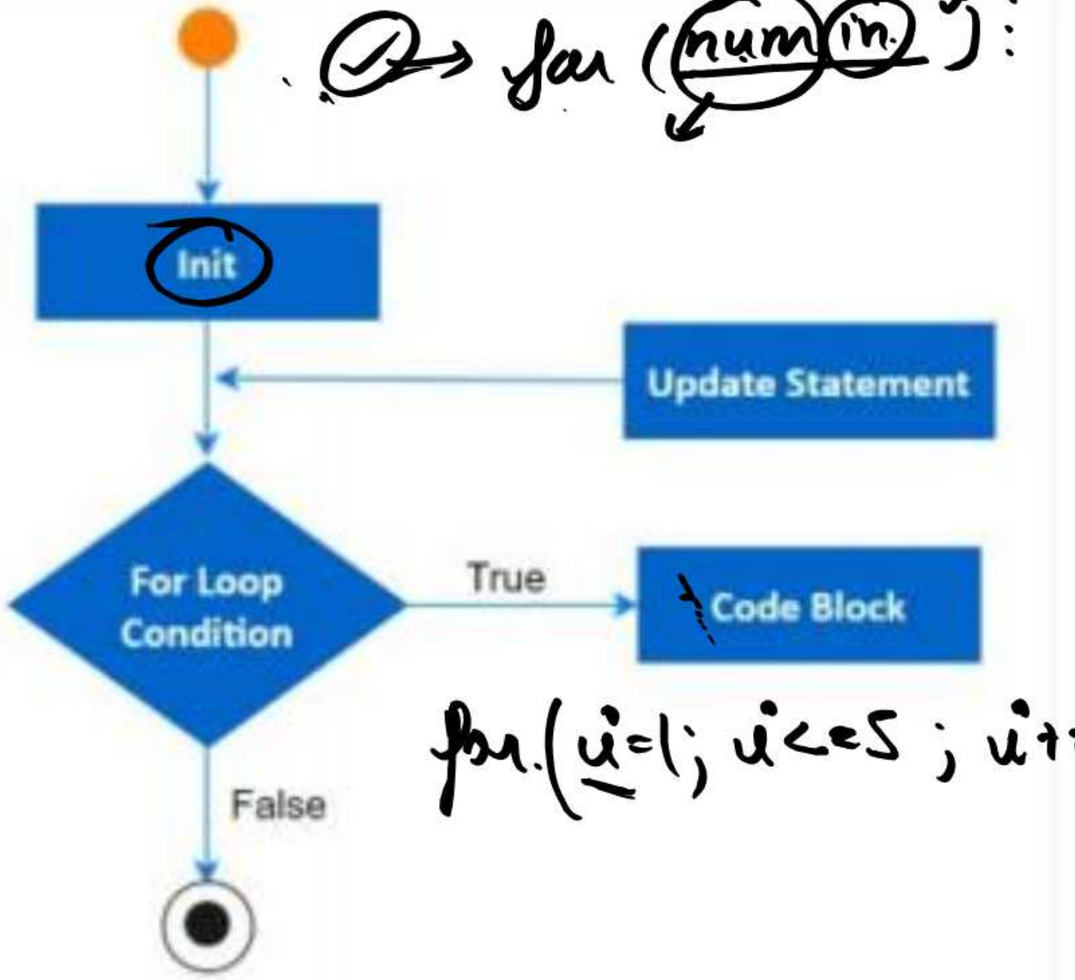
Python for loop

A for loop in Python is used to go through a sequence of items, like a list, a string, or a range of numbers.

It repeats the block of code once for each item in the sequence.

This makes it very useful when you already know how many times you want to loop.

for (num in list):



for (i=1; i<=5; i++)

```
for letter in 'Python': # First Example
 print ('Current Letter :', letter)
fruits = ['guava', 'apple', 'mango']
for fruit in fruits: # Second Example
 print ('Current fruit :', fruit)
print ("Good bye!")
```

Python

guava
apple
mango

Python Nested loop

A nested loop in Python means using one loop inside another loop.

It is commonly used when you need to repeat actions in a multi-level structure, like printing patterns or working with multi-dimensional lists (like 2D arrays).

```
i = 2
while(i < 100):
 j = 2
 while(j <= (i/j)):
   if not(i%j): break
   j = j + 1
 if (j > i/j) : print (i, " is prime")
 i = i + 1
print ("Good bye!")
```

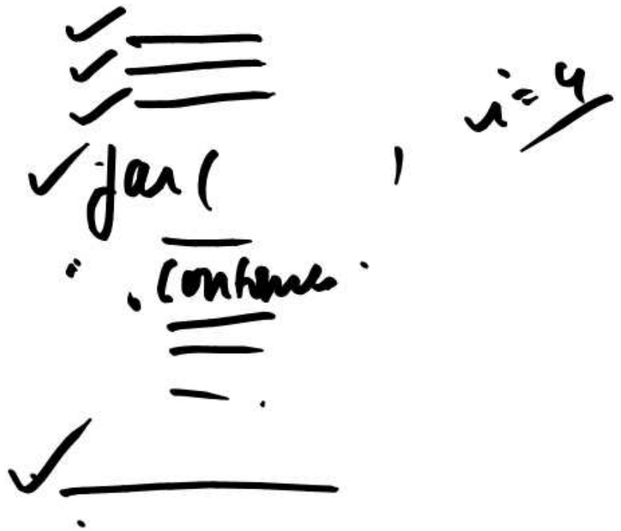
Loop Control Statements in Python

In Python, loop control statements are used to alter the regular flow of a loop.

They enable you to skip iterations, end the loop early, or do nothing at all.

Python's main three loop control statements are:

1-Break
2-Continue
3-Pass



Break

The break statement is used to quickly stop the loop regardless of whether the loop condition is still true.

This signifies that the loop will be terminated and the next statement outside of the loop will be executed.

```
for i in range(10):
  if i == 5:
    break
  print(i)
```

This code in the Python Online Compiler prints the numbers 0 to 4, but then it exits the loop and moves on to the next statement.

Continue

The continue statement is used to skip the current loop iteration and go to the next iteration.
This implies that the code from the current iteration will not be performed, and the loop will continue to iterate until the loop condition is satisfied.

```
for i in range(10):
  if i % 2 == 0:
    continue
  print(i)
```

```
1
3
5
7
9
```

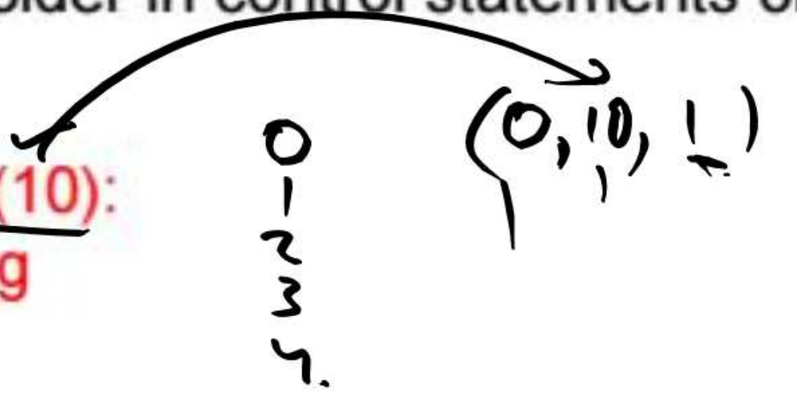
This code generates the odd numbers 1 through 9. Any iterations where the integer is even are skipped by the continue statement.

`break` vs `continue` in Python Loops

Feature	`break`	`continue`
Purpose	Exits the loop **entirely**	Skips the **current iteration** only
Usage	When you want to **stop** the loop early	When you want to **skip** certain steps
Affects	Ends the loop and moves to next statement	Moves to the **next loop iteration**

Pass

The pass statement is a null statement that has no effect. It can be utilized as a placeholder in control statements or empty loops.



```
for i in range(10):
    # Do nothing
    pass
```

This code will not print anything because the pass statement does nothing. It is frequently used as a placeholder in control statements or empty loops.

`for` loop

✅ Use when **you know in advance** how many times you want to iterate (e.g., over a list, range, string, etc.).

`while` loop

✅ Use when you **don't know in advance** how many times to iterate and want to loop **until a condition is false**.

Print Even Numbers from 1 to 20

Initial bound

Upper bound

```
for i in range(2, 21, 2):
print(i)
```

Step

2
4
6
8
10

(2, 21, 2)

2
3

s[0, 7, -] (6)

for(i=2, i<21, i+=2)

for s in 'Explorer':

Sum of Digits of a Number

```
num = 1234
sum_digits = 0
while num > 0:
    digit = num % 10
    sum_digits += digit
    num //= 10
```

num = 123

digit = 4

sum_di = 4

dig = 3

Sum = 7

123 / 10 ⇒ 12.3 ⇒ 12

123 / 11 ⇒ 11.---

123 // 11 ⇒ 11.-- ⇒ 11

Check if a Number is Prime

```
num = 17
is_prime = True

if num < 2:
    is_prime = False
else:
    for i in range(2, int(num**0.5)+1):
        if num % i == 0:
            is_prime = False
            break

print(f"{num} is Prime? {is_prime}")
```

17%

17%2
17%3
17%4
17%5

17%2==
%3==
%4==

%8

$(17)^{1/2} = 4+1$

DSA

Data Structure & Algorithm

⇒ **Complexity** ⇒

→ Time complexity ⇒

→ Space complexity ⇒

* Just by algorithm or pseudocode we can analyse.

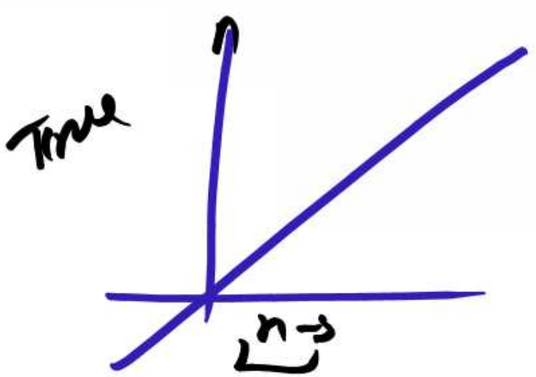
Complexity:

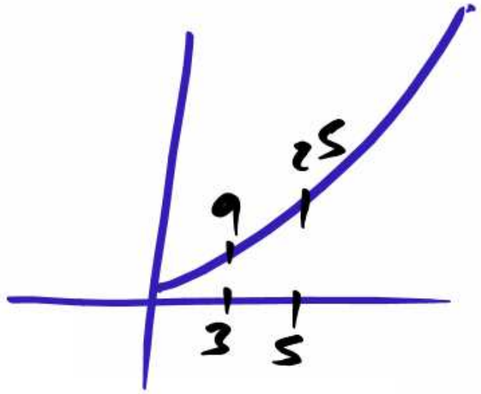
* The mathematical tool used to analyse and describe time complexity is **Asymptotic notations**.

Big(O) notation - (worst case)

Ω (omega) notation (best case)

theta notation (average case)





$$1+2+3+4+5+6\cdots n = \frac{(n)(n+1)}{2}$$

$$1^2+2^2+3^2+4^2+\cdots = \frac{n(n+1)(2n+1)}{6}$$

$$1^3+2^3+3^3+\cdots = \left(\frac{n(n+1)}{2}\right)^2$$

2 mins Summary

TEACHING WALLAH

Topic

Data Structure and Algorithm

TEACHING WALLAH

THANK YOU